IMPERIAL COLLEGE LONDON

EE4-26
EE9-CS5-3
EE9-FPN2-07

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE PART IV: MEng and ACGI

Corrected Copy

ESTIMATION AND FAULT DETECTION

Friday, 10 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : T. Parisini
Second Marker(s) : D. Angeli

© Imperial College London

ESTIMATION AND FAULT DETECTION

Information for candidates:

- **One-step ahead Kalman predictor:**

$$\hat{x}(t+1|t) = F\hat{x}(t|t-1) + K(t)[y(t) - H\hat{x}(t|t-1)]$$

- **Kalman predictor gain**

$$K(t) = FP(t)H^\top \left(V_2 + HP(t)H^\top\right)^{-1}, \quad t = 1,2,\ldots$$

- **Riccati equation**

$$P(t+1) = F\left[P(t) - P(t)H^\top \left(V_2 + HP(t)H^\top\right)^{-1} HP(t)\right]F^\top + V_1, \quad t = 1,2,\ldots$$

- **Realization: observer canonical form**

  Given

$$Y(s)/U(s) = \frac{b_m s^m + \cdots + b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots a_1 s + a_0} \quad \text{with} \quad m < n$$

  then:

$$\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 & -a_{n-2} \\ 0 & \cdots & 0 & 1 & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_m \\ 0 \\ \vdots \\ 0 \end{bmatrix} u \\ y = [0 \cdots 01] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} \end{cases}$$

- **Generic Structure of a Reduced-Order Observer**

$$\begin{cases} \dot{z}_1 = \tilde{A}_{11}z_1 + \tilde{A}_{12}z_2 + \tilde{B}_1 u \\ \dot{z}_2 = \tilde{A}_{21}z_1 + \tilde{A}_{22}z_2 + \tilde{B}_2 u \\ v = z_1 \end{cases}$$

$$\begin{cases} \dot{z} = Fz + Hv + Gu \\ \widehat{x_2} = z + Lv \end{cases} \qquad \begin{cases} L\tilde{A}_{12} - \tilde{A}_{22} = -F \\ L\tilde{A}_{11} + H - \tilde{A}_{21} = FL \\ G + L\tilde{B}_1 - \tilde{B}_2 = 0 \end{cases}$$

1. Consider the closed-loop control system described by the block-scheme shown in Figure 1.1 where $G_1(s)$, $G_2(s)$, $G_3(s)$, and $G_4(s)$ denote the transfer functions of the respective blocks and $K \in \Re$ is a scalar gain.

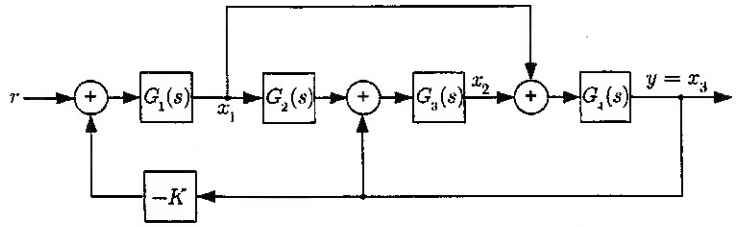


Figure 1.1 Block diagram of the closed-loop control system of Question 1.

a) Determine the transfer function $G_{yr}(s)$ from the input $R(s)$ to the output $Y(s)$ expressed in terms of $G_1(s)$, $G_2(s)$, $G_3(s)$, $G_4(s)$, and K, where $R(s)$ denotes the Laplace transform of the variable $r(t)$ and $Y(s)$ denotes the Laplace transform of the variable $y(t)$.

[3 Marks]

b) Set $G_1(s)$, $G_2(s)$, $G_3(s)$, $G_4(s)$, and K as follows:

$$G_1(s) = \frac{1}{s+5}, \quad G_2(s) = 5, \quad G_3(s) = \frac{1}{s}, \quad G_4(s) = \frac{1}{s+3}, \quad K = 1.$$

Devise a state-space representation of the whole closed-loop control system depicted in Figure 1.1 taking $r(t)$ as the input and $y(t)$ as the output.

[3 Marks]

c) Based on the state-space description of the closed-loop control system obtained in your answer to Question 1-b), show that the systems is not fully observable from the output $y(t)$. Moreover, determine a basis for the non-observable vector subspace.

[3 Marks]

d) Determine the unobservable pole(s) of the closed-loop control system depicted in Figure 1.1 via the analysis of the transfer function from the initial state $x(0^-)$ to the output $y(t)$ (*Hint:*refer to the state-space description obtained in your answer to Question 1-b)).

[4 Marks]

e) Determine a state-space description which is equivalent to that determined in your answer to Question 1-b) and in which the observable and the non-observable sub-systems are clearly identified.

[4 Marks]

f) Remove the setting $K = 1$ given in Question 1-b) and consider a generic choice of $K \in \Re$. Establish conditions on K such that the closed-loop control system depicted in Figure 1.1 is fully observable.

[3 Marks]

2. Consider the electrical circuit depicted in Figure 2.1.

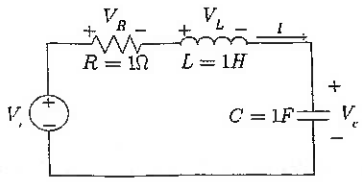


Figure 2.1 Scheme of the electrical circuit of Question 2.

a) The constitutive equations of the electrical components of the circuit depicted in Figure 2.1 are given by:

$$V_R = RI, \quad V_L = L\frac{dI}{dt}, \quad V_c = \frac{1}{C}\int_0^t I d\tau.$$

Devise a state-space representation of the dynamic system given by the electrical circuit depicted in Figure 2.1 taking the voltage $u(t) = V_i(t)$ of the generator as the input and the voltage $y(t) = V_c(t)$ on the capacitor as the output.

[3 Marks]

b) Referring to the state-space representation determined in your answer to Question 2-a), let $x = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^\top$ denote the state vector, let $\hat{x}$ denote the generic state estimate provided by a state observer and by $e = x - \hat{x}$ the corresponding state estimation error. Design a full-order state observer such that the dynamics of the state estimation error satisfies:

$$\dot{e} = Fe$$

for a suitable matrix F whose eigenvalues λ_1, λ_2 of F are $\lambda_1 = -1$ and $\lambda_2 = -1$.

[4 Marks]

c) Using the state-space representation (A,B,C,D) determined in your answer to Question 2-a) and matrix F calculated in your answer to Question 2-b), determine the time-behaviour of the output residual defined as $\varepsilon(t) = y(t) - C\hat{x}(t), \forall t \geq 0$. [5 Marks]

d) Suppose that at some unknown time T_0 an abrupt fault on the generator occurs resulting in a *constant unknown* voltage f that adds up to the known voltage $u(t) = V_i(t)$. Specifically:

$$u(t) = V_i(t), t \leq T_0 \quad \text{and} \quad u(t) = V_i(t) + f, t > T_0.$$

Design a fault detection scheme based on the observer determined in the answer to Question 2-b). Specifically, assuming that $|x(0)| \leq 5$, design a detection threshold $\bar{\varepsilon}(t)$ such that the above actuator fault is detected at some finite time $T_d > T_0$ if the output residual $\varepsilon(T_d)$ satisfies

$$|\varepsilon(T_d)| > \bar{\varepsilon}(T_d).$$

[4 Marks]

e) Consider the abrupt fault described in Question 2-d): devise a method to estimate the fault and discuss the theoretical conditions for this method to yield a correct estimate of the fault. (*Hint:* do not attempt to design the estimator).

[4 Marks]

3. Consider a stochastic dynamic system described by the following difference state equations:

$$\begin{aligned} x(t+1) &= ax(t) + v_1(t), \\ y(t) &= \frac{1}{2}x(t) + v_2(t), \end{aligned} \tag{3.1}$$

where a is a constant scalar and $v_1(\cdot) \sim WGN(0,1)$ and $v_2(\cdot) \sim WGN(0,1)$ are Gaussian zero-mean stochastic processes which are assumed to be mutually independent.

a) Establish conditions on the scalar a such that a one-step ahead optimal steady-state Kalman predictor does exist (*Hint:* the existence of solutions of the Algebraic Riccati Equation (ARE) should be discussed).

[4 Marks]

b) Consider the following two values of the parameter a: $a_1 = \frac{1}{2}$ and $a_2 = -\frac{1}{3}$ and consider the one-step ahead optimal steady-state Kalman predictors for System (3.1) in correspondence of the two values a_1 and a_2 and compute the respective feasible solutions P_1 and P_2 of the ARE.

[3 Marks]

c) Based on matrices P_1 and P_2 obtained in your answer to Question 3-b), compute the corresponding constant gains K_1 and K_2, and then write the difference equations yielding the one-step ahead optimal steady-state Kalman predictions.

[3 Marks]

d) Considering the one-step ahead optimal steady-state Kalman predictor designed in your answer to Question 3-c) for the parameter $a_1 = \frac{1}{2}$, sketch the block diagram of the Kalman predictor with inputs given by $v_1(t)$ and $v_2(t)$ and output given by the estimation error $e(t) = y(t) - \frac{1}{2}\hat{x}(t|t-1)$.

[3 Marks]

e) Using the block-scheme of your answer to Question 3-d), for $a_1 = \frac{1}{2}$ determine the transfer function $G^{(1)}_{v_2 e}(z)$ from the measurement noise $v_2(t)$ to the estimation error $e(t)$. Moreover, for $a_2 = -\frac{1}{3}$ determine the transfer function $G^{(2)}_{v_2 e}(z)$ from the measurement noise $v_2(t)$ to the estimation error $e(t)$. Compare $G^{(1)}_{v_2 e}(z)$ and $G^{(2)}_{v_2 e}(z)$ in terms of the effect of the noise $v_2(t)$ on the estimation error $e(t)$. Discuss your findings.

[7 Marks]

4. Consider a dynamic system described by the following state-space equations:

$$\begin{cases} \dot{x}_1 = -x_1 + x_2, \\ \dot{x}_2 = x_3, \\ \dot{x}_3 = -2x_1 - x_2 + u, \\ y_1 = x_1, \\ y_2 = x_2 + x_3. \end{cases} \tag{4.1}$$

a) Determine the transfer function $G_{yu}(s)$ from the input u to the output $y = [y_1, y_2]^\top$ and, using $G_{yu}(s)$, establish whether System (4.1) is fully observable.

[5 Marks]

b) Determine the order of a reduced-order state observer for System (4.1).

[3 Marks]

c) By a suitable choice of the state vector $x_r(t)$ (with minimum number of states according to your answer to Question 4-b)) and by defining $\widehat{x}_r(t)$ as the estimate of $x_r(t)$, design a reduced-order state observer such that the estimation error $e(t) = x_r(t) - \widehat{x}_r(t)$ obeys the following dynamics:

$$\dot{e} = Fe$$

where all the eigenvalues of F are set to -1 (*Hint*: the design of the reduced-order observer admits more than one solution).

[8 Marks]

d) Sketch the block-diagram of the System (4.1) interconnected with the reduced-order observer designed in your answer to Question 4-c).

[4 Marks]